

Annexure 2B

Manipal Academy of Higher Education, Manipal

Research Methodology course for PhD students in Technology and Management (Online)

(Course Code : RM702)

Credits : 4

1. Research Methodology - By Prof. Edamana Prasad, Prof. Prathap Haridoss, IIT Madras (2 credits)

Archive videos:

<https://nptel.ac.in/courses/121106007>

Week 1: A group discussion on what is research; Overview of research;

Week 2: Literature survey, Experimental skills;

Week 3: Data analysis, modelling skills;

Week 4: Technical writing; Technical Presentations; Creativity in Research

Week 5: Creativity in Research; Group discussion on Ethics in Research

Week 6: Design of Experiments

Week 7: Intellectual Property

Week 8: Department specific research discussions

2. Research Methodology in Natural Sciences by Prof. Soumitro Banerjee | IISER Kolkata (2 credits) <https://nptel.ac.in/courses/127106227>

Week 1: Philosophy of Science (subjective versus objective, materialism versus idealism, causality, etc.)

Week 2: Logical Reasoning (inductive logic, deductive logic, syllogistic logic)

Week 5: Forming a Hypothesis

Week 6: Techniques of Scientific Measurement

Week 7: Testing of hypothesis



Week 8: Methods of Theoretical Research

Week 9: The Art of Scientific Communication

Week 10: Presentation in Seminars and Conferences

Week 11: Sponsored Research

Week 12: Ethical Conduct in Science

Course on Research Writing: Prof. Aradhna Malik

<https://archive.nptel.ac.in/courses/110/105/110105091/>

Course on Academic Writing: Dr Ajay Semalty

<https://ugceresources.in/>

Reference Books:

1. Creswell, J. W., & Creswell, J. D. (2022). Research design (6th ed.). SAGE Publications
2. Saunders, M.N.K., Lewis, P. and Thornhill, A. (2019) Research Methods for Business Students. 8th Edition, Pearson, New York.
3. Duarte, N. (2008). Slide:ology The Art and Science of Presentation. O'Reilly
4. Pranjal Bora, Jibon Saikia & Anil Hazarika (2023). A Concise Book on Research Methodology and Research and Publication Ethics. Notion Press

Assessments:

Assessment pattern for the course shall be as mentioned below:

Assessment 1: MCQ covering all the content of the course (20% weightage); centralized, conducted by the course coordinator

Assessment 2: Develop a 'review of literature section' of protocol (40% weightage) assessed by the PhD supervisor on direction from the coordinator of the Research Methodology coursework

Assessment 3: Develop a protocol (all sections except 'review of literature') as per MAHE template for PhD Protocol Submissions (40% weightage) assessed by the PhD supervisor on direction from the coordinator of the Research Methodology coursework.

